



KARNATAKA ICSE SCHOOLS ASSOCIATION

ICSE STD. X Preparatory Examination 2026

Subject – BIOLOGY

(SCIENCE Paper – 3)

Duration: 2 Hrs

Maximum Marks: 80

Date: 21.01.2026

ANSWER KEY

Answers to this Paper must be written on the paper provided separately.

You will not be allowed to write during first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Section A is compulsory. Attempt any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets []

SECTION A (40 Marks)

(Attempt **all** questions from this Section.)

Question 1

Choose the correct answers to the questions from the given options.

(Do not copy the question, write the correct answers only.)

- (i) (b) Ten
- (ii) (a) Hydathode
- (iii) (c) rr
- (iv) (d) Guanine
- (v) (d) Systole of the left ventricle
- (vi) (d) (A) is false and (R) is true
- (vii) (c) Ion uptake by root hair cells
- (viii) (c) Abscissic acid increases
- (ix) (a) Turgor pressure
- (x) (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (xi) (a) Hepatic vein
- (xii) (d) Axial position of flowers
- (xiii) (a) P – Red blood cells, Q – Platelets
- (xiv) (d) (A) is false and (R) is true
- (xv) (c) Epididymis → Vas deferens → Urethra

[15]

Question 2

- (i) Name the following: [5]
- Antidiuretic hormone (ADH)
 - Sclera
 - Reflex action
 - Glucagon
 - Carbaminohemoglobin
- (ii) Arrange and rewrite the terms in each group in the correct order so as to be in a logical sequence [5]
- Platelets → Thromboplastin → Thrombin → Fibrinogen → Fibrin
 - Photons → Grana → Photolysis → Water molecules → Oxygen
 - Bowman's capsule → Glomerulus → PCT → Loop of Henle → DCT
 - Ovulation → Fertilization → Implantation → Gestation → Parturition.
 - Root hair → Soil water → Cortex → Endodermis → Xylem
- (iii) The diagram of an apparatus given below demonstrates a particular process in plants. Study the same and fill in the blanks with suitable correct answer. [5]
- transpiration
 - uptake
 - water-tight
 - obliquely
 - suction
- (iv) Choose the odd one out from the following terms and name the category to which the others belong: [5]

Odd One Out	Category
(i) Cell wall	Semipermeable membranes
(ii) Seminiferous tubule	Male reproductive accessory glands
(iii) Cochlea	Parts of the eye
(iv) Pesticides	Bio-medical wastes
(v) Acromegaly	Thyroid-related disorders

- (v) Given below is the structure of human ear. Match the structures marked (1) to (6) with their correct function:

Example : (1) – (f) converts air currents into vibration.

(2) – (c) Amplification of vibration

- (3) – (b) Dynamic balance
 (4) – (c) Transmit impulses from ear to brain
 (5) – (d) Hearing
 (6) – (a) equalizes the air pressure

SECTION B (40 Marks)

(Answer **any four** questions from this section)

Question 3

- (i) Diapedesis is the squeezing of leucocytes through the wall of capillaries into the tissues. [1]
 (ii) (a) Osmotic pressure [2]
 (b) Hypotonic solution
 (iii) (a) The mixing up or recombination of genes during meiotic division provides for the innumerable variations and diversity in the progeny. That is how, the children of the same parents, howsoever similar, are different from each other in certain aspects. [2]
 (b) Due to Adrenal Virilism there is an overgrowth of Adrenal Cortex in some women. This causes the development of certain male characteristics like facial hair (beard and moustache) and deep male voice. [2]
 (iv) (a) The Nephron [2]
 (b) Henle's loops and collecting ducts lie in the medulla of kidneys giving it a 'striped' appearance. [3]
 (v) (a) Negative Geotropism
 (b) Auxins
 (c) Shoots show positive Phototropism whereas, roots show positive Geotropism.

Question 4

- (i) Parturition is the act of expelling the full term foetus from the mother's uterus at the end of gestation. [1]
 (ii) In Human body – water reabsorption in kidney tubules. [2]
 In green plants – Opening and closing of stomata. [2]

Tubectomy	Vasectomy
It is the process which involves cutting or ligating of fallopian tubes by nylon thread to close the passage of egg.	It is the process in which vas deferens from each testes is ligated and a small piece between the two ligatures is removed.
It is contraceptive practice in females.	It is contraceptive practice in males.

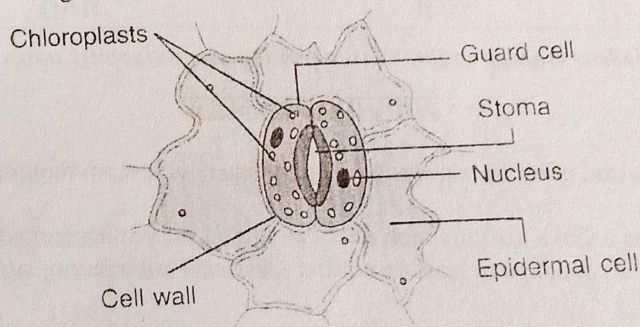
- (a) Decomposers— Fungi/Bacteria [2]
- (b) Respiration by living organisms. [3]

- (a) RrYy
- (b) 9:3:3:1

(c) Law of Purity of gametes (also called the law of Segregation) — The two members of a pair of factors separate during the formation of gametes. They do not blend but segregate or separate into different gametes. The gametes combine together by random fusion at the time of zygote formation.

Question 5

- (i) The embryo respire via diffusion of oxygen from the mother's blood and is used in the oxidation of glucose in the cells with the liberation of energy, but doesn't breathe because its lungs are collapsed and only function after birth. [1]
- (ii) (a) Dwarfism and Gigantism/Acromegaly [2]
(b) Pituitary gland
- (iii) (a) Leaf D would become the most limp. Because it was left uncoated on both surfaces, so it lost the maximum amount of water through transpiration, causes the leaf cells to become flaccid, leading to maximum limpness or wilting. [2]
(b) Leaf C would show the least limping. Because it was coated with vaseline on both surfaces, which blocked all the stomata, which prevents almost all water loss through transpiration. so the leaf cells remained turgid and the leaf stayed fresh, showing minimum limpness.
- (iv) Any two labelling with a diagram. [2]



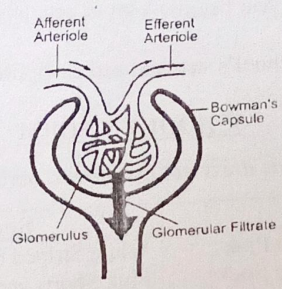
- (v) (a) 1- Placenta, 2- Umbilical cord [3]
- (b) (any one) Acts as a shock absorber, protecting the developing fetus from physical injury while also maintaining stable temperature and pressure and allowing freedom for musculoskeletal development.
- (c) 280 days or 40 weeks

Grey sweat shirt for
For Grade 8 to 10 Lab Coats at
colour as per the school

power of accommodation of the eye is the ability of the eye lens to focus near and far object clearly on the retina by adjusting the thickness of lens. [1]

Menarche	Menopause
Menarche is the first occurrence (onset or beginning) of menstruation in a female.	Menopause is the permanent cessation (stoppage or ending phase) of the menstrual cycle in females.

- (iii) (a) Heart - Right in the centre between the two lungs and above the diaphragm. [2]
(b) Macula lutea - At the back of the eye almost at the centre on the horizontal axis of the eyeball. [2]



(v) (a) [3]

Region	Cyton Location	Axon Location
Cerebrum (Part 1)	In the gray matter (cortex) of the cerebrum	In the white matter (deeper in the cerebrum)(Medulla)
Spinal Cord (Part 2)	In the gray matter (ventral horn for motor neurons, dorsal horn for sensory neurons)(Medulla)	Motor axons exit through the ventral roots, sensory axons enter through the dorsal roots. White matter (Cortex)

(b) Medulla oblongata is a part of brain stem which controls involuntary vital functions like heartbeat, respiration, vasomotor activities and peristalsis. Thus injury to Medulla oblongata disrupts these vital functions and cause sudden death.

(c) Meninges

Question 7

- (i) NADP - Nicotinamide Adenine dinucleotide phosphate [1]
(ii)

Diabetes mellitus	Diabetes insipidus
Insufficient secretion of Insulin hormone	Insufficient secretion of Anti-diuretic hormone (ADH)

- (iii) (a) Chlorophyll is removed by boiling the leaf in alcohol after a brief initial boiling in water to soften it [2]
(b) Chlorophyll is essential for photosynthesis (starch production)

Characters	Australopithecus	Homo sapiens sapiens
Bipedalism	Walked nearly straight	Perfect Bipedalism
Cranial capacity	450 to 600 cm ³	1450 to 1600 cm ³
Development of chin	Lack of chin, prognathous face	Prominent chin, snout disappeared
Forehead and brow ridges	Low forehead; eyebrow ridges projecting over the eyes	Steep forehead; reduced brow ridges; flattened face
Body hair	Body covered with hair	Highly reduced body hair
Height and posture	Nearly upright (3.5 to 4 ft)	Fully erect (up to 6 ft)

(b) *Ramapithecus* → *Australopithecus* → *Homo habilis* → *Homo erectus* → *Neanderthal man* → *Cromagnon* → *Homo sapiens sapiens* (modern man)

(a) Hyperopia (Hypermetropia)

(b) Shortening of the eyeball from front to back or the lens is too flat (*any one*)

(c) A convex lens is used to correct this defect.

Question 8

(i) Population density is the number of individuals per square kilometre (km²) at any given time.

(ii) (a) Correct order is:

(3) Stimulus → (2) Detection by receptors → (1) Co-ordination by brain → (5) Effector change →

(4) Response

(b)

Voluntary action	Reflex action
It is performed under the control of the brain and requires conscious thought (e.g., picking up a book).	It occurs automatically and does not involve conscious control, usually for quick protection (e.g., withdrawing hand from a hot object).

(iii) (a) Erosion of ancient monuments, statues and sculpture by the acid reacting with the calcium.

(b) Degrades the visibility and skin diseases.

(iv)

Structure	Location
Corpus callosum	Brain (between the two cerebral hemispheres)
Corpus luteum	Ovary (forms from ruptured follicle after ovulation)

$Q_1 = 4$
 $Q_2 = 9$

- (a) Ventricular systole and Atrial diastole. Because both the semi-lunar valves are open. [3]
- (b) Part 1 (Pulmonary artery) – Deoxygenated blood; Part 2 (Aorta) – Oxygenated blood
- (c) The left ventricle has thicker walls because it has to pump blood to the entire body, which requires higher pressure. Whereas the right ventricle only pumps blood to the lungs, which is a shorter distance and needs less pressure.
